Transcription & Annotation Standards

The conventions that govern every gold reference in this project — what gets transcribed, how it's written, and what happens to it afterward.

This is the **policy document**. For the step-by-step ELAN walkthrough, see annotator_guide.md. For how these standards meet the ASR benchmark, see asr_provider_spec.md.

> **Scope:** we benchmark on *our* audio — OSCE simulated-patient encounters — not public datasets. Generic WER numbers don't transfer to this domain. That's the entire reason this document exists.

> **Version this file.** Inconsistent conventions silently inflate WER and silently destroy inter-annotator agreement. Every change here invalidates comparisons across the boundary; see §12.

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1. The governing principle

> **Every convention must be machine-removable without a human decision.**

This is what separates a markup scheme you can strip back to a clean transcript from one you're stuck with forever. It's why prosody, emphasis, and speech-rate notation are out of scope no matter how interesting they are: they can't be removed mechanically, they have poor agreement, and no metric consumes them.

It also means the "clean transcript for other users" is a **derived artifact**, never a separately-maintained file. One source, three renderings (§7).

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2. What gold covers

Layer	For metric	Output
Verbatim transcript	WER / CER	`<case_id>.gold.txt`
Speaker turns	DER / JER	`<case_id>.gold.rttm`
Word timing	boundary error	`<case_id>.gold.ctm` — *not built*

Annotators produce the first two from a single ELAN file. **Nobody times individual words by hand.** WER ignores timing entirely; DER needs turn boundaries, not word boundaries. Word-level gold is a separate, expensive layer worth building only if alignment accuracy becomes a research question in itself.

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3. Sampling

You will not hand-annotate 300–400 encounters. Draw a **stratified sample** (start 5–10 cases) spanning the conditions that actually break ASR:

- quiet vs. noisy rooms
- number of speakers and how often they overlap

- learner accent and speech rate
- audio source (which camera mic)

Record why each case was chosen. Report results as estimates over this sample with the strata named — never imply the whole corpus is gold.

Excerpts are valid references

A reference does **not** have to be a whole encounter. A carefully annotated 5-minute excerpt is a valid reference as long as scoring is restricted to that span — WER is a rate, not a total.

This matters because annotator time is the binding constraint. For a fixed budget, **six 5-minute excerpts across varied conditions beat two full 20-minute encounters**: better strata coverage, more cases, same cost. Prefer breadth over completeness unless a specific question needs whole-encounter continuity.

Record the span bounds with the reference, so scoring can restrict to it and so nobody later mistakes an excerpt for a full transcript.

Constraint: gold annotation is human-local and could be done on any case, but cross-provider scoring requires a case that's cloud-releasable (`cases.cloud_release`, provider spec §9). Draw the sample from releasable cases only, so every reference scores every engine. Today that's no constraint at all — the corpus is uniformly OSCE — but note it before the pool stops being uniform.

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4. Tiers

Six tiers, pre-created in the template. Annotators **never create tiers** except for a genuinely unanticipated party.

Tier	Holds	In `gold.txt`?	In `gold.rttm`?
`LEARNER`	the student	✅	✅
`PATIENT`	the standardized patient	✅	✅
`PRECEPTOR`	the supervising clinician	✅	✅
`ANNOUNCEMENT`	PA / station timing calls	❌	✅
`OUTSIDE_ROOM`	intelligible hallway bleed	❌	✅
`NOTES`	annotator observations	❌	❌

Why `ANNOUNCEMENT` and `OUTSIDE_ROOM` are in the RTTM but not the transcript: the benchmark should measure transcription of the *encounter*, not whether an engine caught a stranger in the hallway. But a diarizer will detect those voices as acoustically distinct regardless of what gold says — omitting them from the RTTM entirely would falsely penalize a diarizer via DER for correctly noticing a real extra voice. Transcribe them (§5's verbatim rule applies — don't silently drop real audio), then exclude the text and keep the boundaries.

`NOTES` is sparse and non-exhaustive. Free text, no timing precision required. It exists to answer "why did this case score badly?" — noise bursts, very quiet stretches, heavy crosstalk, unfamiliar accents,

equipment problems. It feeds error analysis and sampling rationale at near-zero cost.

> **There is no noise tier, and no exhaustive event tier.** Noise enters neither metric, and the scorer's bracket-stripping rule (§7) already makes provider audio-event tags scoring-neutral. Exhaustive noise annotation is the classic scope creep that stalls annotation projects. `NOTES` serves the real need.

Two things the tier scheme gives you for free

Overlap needs no markup. Annotators segment each role tier in its own listening pass, so overlapping speech is simply two tiers carrying annotations at the same timestamps. It's emergent from the geometry.

There is no `SPEAKER_NN` intermediate. Annotators segment directly onto role tiers, because a human can hear who is talking. Gold RTTM labels are role names; DER solves optimal label assignment anyway, so a diarizer's `SPEAKER_00` maps cleanly at scoring time.

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5. Segmentation

Unit: the utterance / breath group — a natural phrase boundary, typically a few seconds. Not the full speaker turn.

A 90-second history-taking summary as one annotation is unwieldy to type, to review, and to compare between annotators. Utterance-level segments are far easier to work in ELAN's Transcription Mode and give finer-grained inter-annotator comparison.

Turn-level RTTM is produced by **mechanically merging adjacent same-tier segments** at export. Deterministic, costs nothing.

Boundary precision: bound the speech, don't chase it. Roughly a tenth of a second is plenty. Nothing downstream needs better, and precision you can't reproduce is worse than none.

Silent pauses are never marked. Utterance timestamps already measure them in milliseconds — continuous, not categorical, and free. Pause duration is the strongest hesitation signal in clinical communication research and you get it without a single keystroke.

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6. Transcription conventions

Verbatim, not cleaned. Transcribe what was said. False starts, repetitions, grammatical errors, and self-corrections all stay. The cleaned rendering is derived later (§7) — cleaning at the keyboard is irreversible.

Filled pauses — semantically empty, removed at L1

`uh` · `um` · `hmm`

> **`er` and `erm` are deliberately excluded.** They're British-convention variants, and `uh` vs `er` is perceptually a coin flip for American speakers — pure inter-annotator noise for no analytic gain. Write `uh`.

Backchannels — carry meaning, never removed at any level

Token	Means
`mm-hmm`	yes
`uh-huh`	yes
`uh-uh`	no
`mm-mm`	no
`huh?`	request for repetition

> ⚠ **`uh-huh` and `uh-uh` are one letter apart with opposite meanings.** A patient answering "uh-uh" to "any chest pain?" is a clinical *no*. This is a genuine annotator hazard and ASR providers confuse the pair constantly — expect "provider flips patient yes/no backchannels" to become a real finding.
>
> **Discriminator:** `uh-uh` has a glottal stop between syllables and falls then rises. `uh-huh` has an audible /h/ and rises. **If you can't tell, listen to what comes next** — the other speaker's response disambiguates almost every time. Still unsure? Write `[unintelligible: uh-huh?]`. A flagged unknown is recoverable; a flipped clinical *no* is not.

Prolongation — trailing colon

`so:` · `the:` · `I: think`

One colon regardless of length. Never grade duration — that destroys agreement for information already measurable from the waveform.

Fragments and cutoffs — trailing hyphen

`par- particular` · `I was gon-` · `I went- I drove`

The strongest stammer and self-repair signal, and standard LDC/Switchboard convention. Self-repair needs no additional markup — the hyphen already marks the abandonment.

Repetitions — no markup at all

Just transcribe them: `I I I went`. Adjacent identical tokens are mechanically detectable, so L2 collapses them automatically. Zero annotator burden, full recoverability.

Annotator uncertainty

Tag	Use
`[unintelligible]`	nothing recoverable
`[unintelligible: pressure?]`	a best guess worth preserving

Both are stripped from *both* sides before scoring (§7), so uncertain spans never count for or against an engine. Use them freely — guessing is the expensive mistake.

Vocalizations — optional, non-exhaustive

`[laugh]` · `[cough]` · `[sigh]` · `[throat clear]`

Tag them when clinically interesting; skip them otherwise. Scoring-neutral either way.

Numbers and units — as spoken

`one twenty over eighty`, not `120/80`. `fifteen milligrams`, not `15 mg`.

Providers' number formatting varies wildly; canonicalizing both sides at scoring time measures the recognizer instead of the vendor's formatter. Writing digits at the keyboard throws away what was actually said.

Domain terminology — spell from the glossary

`benchmarks/configs/glossary.yaml` is the canonical spelling list. It is **the same file** the ASR providers get biased toward (provider spec §3). Hit a term that isn't in it? Add it — that's how the glossary grows.

Casing and punctuation — annotate naturally

The scorer normalizes both away. Punctuate for readability and move on; don't agonize.

Explicitly out of scope

Prosody · emphasis · intonation contours · overlap markers · speech rate.

Overlap is already emergent from tier geometry (§4). The rest violate §1 — they can't be stripped mechanically, they have poor agreement, and nothing consumes them.

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7. Normalization levels

One source, three renderings. Each is a deterministic transform of the one above.

Level	Contents	Purpose
L0 verbatim	everything in §6	the gold file; **immutable**
L1 filler-neutral	L0 – filled pauses, prolongation marks, bracketed spans	second WER number
L2 clean	L1 – fragments, – collapsed repetitions	readable transcript for others

L0 and L1 are scoring profiles. L2 is a deliverable, never a metric.

Shared transforms applied identically to reference and hypothesis: casing, punctuation, whitespace, **bracket stripping**, name redaction (§8), number canonicalization.

> **Known limitation:** L2 drops the fragment token but *not* the whole abandoned phrase — `I went- I drove` becomes `I went I drove`, not `I drove`. Removing abandoned material properly

> requires a judgment call about where the repair started, which violates §1. Stated honestly rather than approximated badly.

A research affordance

L0 minus L1 is a disfluency measure. Filled-pause rate, fragment rate, and pause duration per learner turn fall straight out of the gold files with no additional annotation — so learner fluency becomes analyzable as a by-product of building ASR references.

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8. Names and de-identification

Annotators type names verbatim, exactly as heard. They make no redaction decisions at the keyboard.

This is deliberate, and it reverses the intuitive approach. The reasoning: **the hypothesis transcripts contain the real names regardless** — the provider returns "Jamie" and that response lands on disk. Redacting only the gold file protects nothing and creates a reference/hypothesis mismatch that scores as an error against a provider that was *correct*. And fixing that requires a name list anyway. Since the list is needed either way, use it once, mechanically.

Two layers

Layer 1 — protected working layer (`data/`, gitignored): ELAN files with real names, raw provider responses, normalized transcripts. Scoring happens here.

Layer 2 — publishable layer (`benchmarks/references/`, tracked): the de-identified gold export, produced **mechanically** from Layer 1. Never hand-edited.

The redaction map is **already in the manifest** — `cases.learner_name`, `cases.sp_name` (and `preceptor_name`, to be added). No separate name file.

Redaction is applied to both reference and hypothesis before scoring, so the reported number belongs to text you can actually publish.

Why this is better: redaction becomes a logged deterministic transform rather than an unauditable human judgment made mid-flow; Layer 2 is derived, so it can be regenerated; and annotators just transcribe.

> **Handling caution:** Layer 1 ELAN files are the most sensitive artifacts an annotator touches. They stay on managed storage, are never emailed, and never enter git.

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9. Process — blind first

Annotation is **from scratch**. No machine draft is corrected, ever.

Correcting a Whisper draft biases gold *toward Whisper*: annotators accept plausible-looking errors, flattering the very engine under test. The bias is invisible in the output and fatal to the benchmark's purpose.

Pass 1 — blind

The annotator works from the template and the audio alone. No draft, no existing transcript, no other annotator's work. Output is `<case_id>.pass1.eaf`, which is **frozen and immutable** on completion.

Pass 2 — adjudication

Only after pass 1 is frozen may the machine draft be consulted. Divergences flag spots to **re-listen** — never to accept. The reviewer's job is to decide what the audio actually contains, not to reconcile two documents. Output is `<case_id>.gold.eaf`.

Because pass 1 is preserved, the human-vs-machine divergence rate is a free signal: a systematic pattern in it usually means either a convention gap (§12) or a genuine engine weakness worth reporting.

Never let a pipeline re-run overwrite human work. Gold files carry distinct names and are treated as immutable.

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10. Quality control

Double-annotate a subset (target ~20%) fully independently — two blind pass 1s on the same case, neither annotator seeing the other's work.

Measure agreement: WER *between annotators* for transcript, DER between their speaker labels. Report at both L0 and L1 — L0 disagreement concentrated in disfluency markup means a convention problem, not a perception problem.

High disagreement means §6 is underspecified. Fix the standard and re-annotate. Do not average the confusion.

> **The annotator-vs-annotator error rate is your noise floor.** No engine can meaningfully "beat" it. Report it alongside every engine WER — a benchmark without it can't distinguish a good model from a saturated metric.

When there's only one annotator

With a single annotator there is **no noise floor at all**. That's a realistic situation here, so name the options rather than letting it slide:

1. **Second-annotate one excerpt yourself.** Five minutes is enough for *an* estimate, and a rough floor beats none.
2. **Report engine numbers with an explicit statement that no floor was established.**

Do not quietly omit it. The absence of a floor is a limitation on every number in the report, and it belongs next to them.

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11. File layout

```
elan/
└── template.etf                          # the versioned annotation template
```

```
data/gold/<case_id>/                    # GITIGNORED — Layer 1, real names
├── <case_id>.pass1.eaf                 # frozen blind pass
└── <case_id>.gold.eaf                  # adjudicated

benchmarks/
├── configs/
│   ├── glossary.yaml                   # canonical spellings + provider
biasing
│   └── providers.yaml                  # capability registry
├── references/                         # TRACKED — Layer 2, de-identified
│   ├── <case_id>.gold.txt
│   └── <case_id>.gold.rttm
├── runs/                               # GITIGNORED — per-run outputs
└── results/
    └── runs.csv                        # TRACKED — append-only,
provenance-stamped
```

> **The template is tracked in git** (`elan/template.etf`), not under `data/` — it's a standard, not data, and an untracked template gives no guarantee that two annotators are using the same tier set. Changing its tiers is a standards change (§12), not a file edit.
>
> The older duplicate at `data/template.etf` (leftover `default` tier, no `NOTES` tier) was deleted 2026-08-26. If it ever reappears — e.g. someone's local checkout regenerates it — delete it again; two competing templates is exactly the ambiguity this layout removes.

Current gaps

- **No `.eaf → gold` exporter exists.** Under the from-scratch workflow this is no longer optional — it's the only path from annotator output to `gold.txt` + `gold.rttm`. It must apply tier exclusion (§4), merge adjacent same-tier segments (§5), and redact from the manifest (§8).
- **DER scoring is not implemented** in `run_benchmark.py` — it computes WER/CER only. Collecting the RTTM layer is not wasted (§5 produces exactly what DER needs), but don't expect a DER number until that path exists.

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12. Change control

Changing §6 or §7 **invalidates comparisons across the change**. So:

1. Every change gets a version bump and a dated entry below.
2. Gold files record the standards version they were annotated under.
3. `normalizer_version` in the benchmark provenance moves with §7.
4. Existing gold is **not** silently re-interpreted under new rules — either migrate it deliberately and say so, or scope results to a version.

The most common trigger is §10 finding disagreement that traces to an underspecified rule. That's the system working: fix the rule, bump, re-annotate the affected subset.

Version	Date	Change
1.0	2026-08-26	Initial standard. From-scratch blind annotation, utterance-level segmentation, L0/L1/L2 levels, verbatim names with mechanical redaction. Supersedes `gold_annotation_guide.md`.